

The following problems will help you to prepare for the final exam. They indicate the style (but not the amount!) of problems that will form the bulk of the final: 1. giving examples, 2. stating/correcting important definitions, 3. stating/correcting and illustrating important results, 4. proving smaller results, 5. ASN, 6-10. free response questions, mostly similar in content, style, and difficulty to the HWs and Homeworks. You should also be prepared for questions like those on the Midterm Review Problems and from the midterm itself; you should also be prepared to use all of the skills from the skills quizzes. A key for these review problems will be posted in Files.

1. Give an example of each of the following, or explain why it is not possible.

a) A nilpotent matrix of rank 2.

$$\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

b) An orthogonal matrix that is nilpotent.

Not possible. Orthogonal matrices are invertible; nilpotent ones aren't. For instance, a nilpotent matrix has a non-trivial nullspace and has determinant 0.

c) A pair of vectors in $\mathbb{R}^3$ such that their span is a plane.

$$\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \text{ and } \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

d) A pair of orthogonal vectors in $\mathbb{R}^3$ such that their span is a plane.

$$\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \text{ and } \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

e) A pair of vectors in $\mathbb{R}^3$ such that their span is a line.

$$\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \text{ and } \begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix}$$

f) A pair of orthogonal vectors in $\mathbb{R}^3$ such that their span is a line.

$$\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \text{ and } \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

g) An orthonormal basis for the nullspace of $\begin{pmatrix} 4 & 6 & 10 \end{pmatrix}$.

$$\frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} \text{ and } \frac{1}{\sqrt{114}} \begin{pmatrix} 8 \\ -7 \\ 1 \end{pmatrix}$$

h) A non-zero matrix satisfying the polynomial equation $x^4 + x = 0$.

$$x^4 + x = x(x^3 + 1) = x(x + 1)(x^2 - x + 1)$$

So $-I$ satisfies the equation, for any dimension.

i) An invertible matrix satisfying the polynomial equation $x^4 + x = 0$.

Again, $-I$ satisfies the equation, and $-I$ is invertible.

j) A non-zero matrix satisfying the polynomial equation $x^4 = 0$.

Any nilpotent matrix of dimension at most 4 will work, e.g.

$$\begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

k) An orthogonal matrix satisfying the polynomial equation $x^4 = 0$.

Not possible. Any matrix satisfying the equation $x^4 = 0$ is nilpotent, but orthogonal matrices are invertible.

l) An invertible matrix satisfying the polynomial equation $x^4 = 0$.

Not possible, same issue.

m) Two non-equal matrices, both with characteristic polynomial $\lambda^2 - 5\lambda + 6$.

$\lambda^2 - 5\lambda + 6$ factors as $(\lambda - 2)(\lambda - 3)$. The following two matrices are not equal but have this as their characteristic polynomial. (Note: while they are not equal, they *are* similar.)

$$\begin{pmatrix} 3 & 1 \\ 0 & 2 \end{pmatrix} \text{ and } \begin{pmatrix} 3 & 0 \\ 0 & 2 \end{pmatrix}$$

n) A matrix that is diagonal but not diagonalizable.

Not possible. Every diagonal matrix is diagonalizable, as every matrix is similar to itself: $A = IAI^{-1}$.

o) A matrix with 4 distinct eigenvalues.

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 4 \end{pmatrix}$$

has eigenvalues 1, 2, 3, and 4

p) An orthogonal matrix that is not invertible.

Not possible. Every orthogonal matrix is invertible: its inverse is its transpose.

q) An inner product on $\mathbb{R}^2$ that is not the standard inner product.

For instance: $\langle x, y \rangle = 2x_1y_1 + 3x_2y_2$. (The usual one is $x_1y_1 + x_2y_2$.)

r) Two vectors such that $\|v\| = \|w\| = 1$ and $v \cdot w = 1/2$.

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} \text{ and } \begin{pmatrix} 1/2 \\ \sqrt{3}/2 \end{pmatrix}$$

s) Two vectors such that $\|v\| = \|w\| = 1$ and $v \cdot w = 2$.

Not possible. By Cauchy–Schwarz, the absolute value of the inner product of two vectors is at most the product of their norms, which in this case is 1.

t) A square matrix with eigenvalues 1, 2, and 3.

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$$

u) A non-square matrix with eigenvalues 1, 2, and 3.

Not possible. Non-square matrices do not have eigenvalues.

v) A square matrix with singular values 1, 2, and 3.

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix} \text{ again works, since calling it } A \text{ we have}$$

$A^T A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 9 \end{pmatrix}$ and the (positive) square roots of this matrix's eigenvalues are the singular values of A . Therefore we get back 1, 2, 3.

w) A non-square matrix with singular values 1, 2, and 3.

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \\ 0 & 0 & 0 \end{pmatrix} \text{ works, since } A^T A \text{ again equals } \begin{pmatrix} 1 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 9 \end{pmatrix}$$

x) A non-diagonal matrix similar to a diagonal matrix.

For instance, $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ is similar to $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

y) A non-identity matrix similar to an identity matrix.

Not possible. The only matrix similar to the identity matrix is itself: $PIP^{-1} = PP^{-1} = I$.

z) A non-orthogonal matrix similar to an orthogonal matrix.

$\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$ is orthogonal, but $\begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} -1 & -2 \\ 0 & 1 \end{pmatrix}$ is not, and they're similar.

aa) An orthogonal matrix with determinant -1.

$$\begin{vmatrix} -1 & 0 \\ 0 & 1 \end{vmatrix} = -1$$

bb) An orthogonal matrix with determinant 10.

Not possible. An orthogonal matrix takes the standard basis to some orthonormal basis; and so the signed volume determined by this new basis must be 1 or -1, and so therefore also the

determinant of the matrix.

cc) An upper triangular matrix that is not a Jordan block.

$$\begin{pmatrix} 1 & 5 \\ 0 & 1 \end{pmatrix}$$

dd) An invertible matrix that is a Jordan block.

For instance, $\begin{pmatrix} 3 & 1 \\ 0 & 3 \end{pmatrix}$ is a 2×2 Jordan block with $\lambda = 3$. It is invertible since its determinant is 9 and so nonzero.

ee) A unitary matrix and a non-unitary matrix that are row equivalent.

$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ are row equivalent, but only the latter is unitary (and even orthogonal).

ff) A matrix with eigenvalues 4 and $2 + i$.

$$\begin{pmatrix} 4 & 0 \\ 0 & 2 + i \end{pmatrix}$$

gg) A real matrix with spectrum that contains 4 and $2 + i$.

Since the problem doesn't specify that these are the *only* eigenvalues of the matrix, we can

have: $\begin{pmatrix} 4 & 0 & 0 \\ 0 & 0 & \sqrt{5} \\ 0 & -\sqrt{5} & 0 \end{pmatrix}$. It is a real matrix with characteristic polynomial $(4 - \lambda)(\lambda^2 + 5) = (4 - \lambda)(\lambda - 2 - i)(\lambda - 2 + i)$, and so has 4, $2 + i$, and $2 - i$ as eigenvalues.

gg') A real matrix with spectrum that equals 4 and $2 + i$.

This is not possible, since 4 and $2 + i$ cannot be the only eigenvalues of a real matrix. A real matrix has a characteristic polynomial with real coefficients, and the non-real roots of a real polynomial come in complex conjugate pairs: if $2 + i$ is an eigenvalue of a real matrix, so is $2 - i$.

hh) A matrix with spectrum that contains 5 and 6 and that has determinant 20.

$$\begin{pmatrix} 5 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 2/3 \end{pmatrix}$$

ii) A matrix that has only 5 and 6 as eigenvalues and that has determinant 20.

Not possible. The product of a matrix's eigenvalues is its determinant, and no product of 5's and 6's equals 20.

jj) A matrix that has only 2 and 5 as eigenvalues and that has determinant 20.

$$\begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 5 \end{pmatrix}$$

kk) A nilpotent matrix with determinant 9.

Not possible. The determinant of every nilpotent matrix is 0; it has a non-trivial nullspace.

ll) An invertible matrix with determinant 9.

(9) or $\begin{pmatrix} 3 & 1 \\ 0 & 3 \end{pmatrix}$ or many others.

mm) An orthogonal matrix with determinant 9.

Not possible. The determinant of an orthogonal matrix is either 1 or -1, depending on whether it preserves or reverses orientation.

nn) An orthogonal matrix with eigenvalues 3 and 1/3.

Not possible. Every real eigenvalue of an orthogonal matrix must be 1 or -1, since orthogonal matrices preserve lengths of vectors

oo) A 2×2 matrix that is not in Jordan canonical form.

$\begin{pmatrix} 3 & 1 \\ 0 & 2 \end{pmatrix}$; its JCF is $\begin{pmatrix} 3 & 0 \\ 0 & 2 \end{pmatrix}$, as it is diagonalizable.

pp) Two matrices, each with nullity 2, whose product has nullity 3.

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

qq) Two matrices, each with nullity 2, whose product has nullity 4.

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

rr) Two matrices, each with nullity 2, whose product has nullity 5.

Not possible. The maximum nullity of a product is the sum of the nullities of the factors.

ss) A matrix with 2 real and 2 non-real eigenvalues.

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & i & 0 \\ 0 & 0 & 0 & 2i \end{pmatrix}; \text{ or for an example of a real matrix with this property: } \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

tt) A real matrix with no real eigenvalues.

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

uu) A complex matrix with no complex eigenvalues.

Not possible. Every complex $n \times n$ matrix has a complex polynomial of degree n as its characteristic polynomial, and the Fundamental Theorem of Algebra implies that every such polynomial has n complex roots, counted with multiplicity. Also, note that the real numbers are contained in the set of complex numbers!

vv) A real number that is never the eigenvalue of a matrix.

Not possible. Every real number is the eigenvalues of some matrix. For instance, $k \in \mathbb{R}$ is an eigenvalue of the matrix $\begin{pmatrix} k & 0 \\ 0 & k \end{pmatrix}$.

ww) A real matrix that is not similar to a real upper triangular matrix.

$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. Its eigenvalues are i and $-i$; in upper triangular form, these would appear along the diagonal, but then the matrix is not real.

xx) A complex matrix that is not similar to a complex upper triangular matrix.

Not possible. Every complex matrix is similar to a complex upper triangular matrix. This is implied by the Schur representation.

yy) A matrix with trace 0 and determinant 1.

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

zz) A matrix with trace 0 and eigenvalues 2 and 3.

$\begin{pmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & -5 \end{pmatrix}$. If 2 and 3 were the only eigenvalues of the matrix, then it would not be possible, since the sum of the eigenvalues is the trace, and no sum of 2's and 3's equals 0.

A) A square matrix A for which there exists no square matrix B where $B^2 = A$.

Consider the nilpotent matrix $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$. We have that $A^2 = 0$. In fact, it follows from the Cayley–Hamilton theorem that the square of every nilpotent 2×2 matrix equals 0. If there were a matrix B such that $B^2 = A$, then B would have to be nilpotent, since $B^4 = A^2 = 0$. But then we have a contradiction: $B^2 = A \neq 0$, and also $B^2 = 0$ by the Cayley–Hamilton fact above.

B) Two matrices A and B that commute but whose squares A^2 and B^2 do not commute.

Not possible. If A and B commute, we have $A^2B^2 = AAB B = ABAB = BAAB = BABA = BBAA = B^2A^2$, and so A^2 and B^2 also commute.

C) Two matrices A and B that do not commute but whose squares A^2 and B^2 commute.

There are many examples. For instance, $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, $B = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$. Other examples include reflections in any two lines through the origin in $\mathbb{R}^2$ that are not perpendicular. Their squares commute because they are both the identity. But the matrices themselves don't because they are rotations about the origin of the same amount but in opposite directions.

D) An orthonormal basis for $\mathbb{R}^2$ that contains no standard basis vectors.

$$\left\{ \begin{pmatrix} 3/5 \\ 4/5 \end{pmatrix}, \begin{pmatrix} -4/5 \\ 3/5 \end{pmatrix} \right\} \text{ or } \left\{ \begin{pmatrix} -1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ -1 \end{pmatrix} \right\}$$

E) An orthonormal basis for $\mathbb{R}^3$ that contains exactly two standard basis vectors.

$$\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \text{ and } \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \text{ and } \begin{pmatrix} 0 \\ 0 \\ -1 \end{pmatrix}$$

F) A nilpotent permutation matrix.

Not possible. Every permutation matrix is invertible, and no nilpotent matrix is invertible.

G) An invertible permutation matrix.

Any permutation matrix works.

H) A permutation matrix with trace 0.

Any permutation matrix with all 0's on the diagonal works; this is the same as the permutation having no fixed points. For example:

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

I) A basis for the subspace of $M_{2 \times 2}^{\mathbb{R}}$ consisting of symmetric matrices.

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \text{ and } \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \text{ and } \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

The idea is that a 2×2 symmetric matrix can have arbitrary values along its diagonal, but the upper right and lower left entries have to match.

J) A matrix with an eigenvalue λ with algebraic multiplicity 4 and geometric multiplicity 2.

$$\begin{pmatrix} 5 & 1 & 0 & 0 \\ 0 & 5 & 0 & 0 \\ 0 & 0 & 5 & 1 \\ 0 & 0 & 0 & 5 \end{pmatrix}$$

K) A matrix with an eigenvalue λ with algebraic multiplicity 4 and geometric multiplicity 4.

$$\begin{pmatrix} 5 & 0 & 0 & 0 \\ 0 & 5 & 0 & 0 \\ 0 & 0 & 5 & 0 \\ 0 & 0 & 0 & 5 \end{pmatrix}$$

L) A matrix with an eigenvalue λ with algebraic multiplicity 4 and geometric multiplicity 0.

No possible. Whenever a matrix has an particular eigenvalue, it always has a corresponding eigenspace of dimension at least 1.

M) Two matrices with different ranks but the same nullspace.

Two matrices with the same nullspace have the same domain and so the same number of columns. By the Rank-Nullity theorem, they each have this same value of their number of columns as the sum of their respective ranks and nullities. Since they have the same nullspace, they have equal nullities, and therefore also equal ranks.

N) Two matrices with different ranks but the same nullity.

$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$ has rank 4 and nullity 0. $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ has rank 3 and nullity 0.

O) Two square matrices with different ranks but the same nullity.

The same answer as in (N) works.

P) A square matrix A and its $D + N$ decomposition.

A nice special case is when A is upper triangular and all of its eigenvalues are equal:

$$\begin{pmatrix} 3 & 1 & 2 \\ 0 & 3 & 3 \\ 0 & 0 & 3 \end{pmatrix} = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{pmatrix} + \begin{pmatrix} 0 & 1 & 2 \\ 0 & 0 & 3 \\ 0 & 0 & 0 \end{pmatrix}$$

Note that for the next few problems, ideas from diagonalization could be used instead; here we give another approach.

Q) A matrix that has $\begin{pmatrix} 2 \\ 3 \end{pmatrix}$ as an eigenvector.

$$\begin{pmatrix} 2 & 2 \\ 3 & 3 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 10 \\ 15 \end{pmatrix}$$

R) A matrix that has $\begin{pmatrix} 2 \\ 3 \end{pmatrix}$ as an eigenvector with eigenvalue -1.

$$\begin{pmatrix} 2 & -2 \\ 3 & -3 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} -2 \\ -3 \end{pmatrix}$$

S) A non-diagonal matrix that has $\begin{pmatrix} 2 \\ 3 \end{pmatrix}$ as an eigenvector with eigenvalue -1.

$$\begin{pmatrix} 2 & -2 \\ 3 & -3 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} -2 \\ -3 \end{pmatrix}$$

T) A matrix that has $\begin{pmatrix} 2 \\ 3 \end{pmatrix}$ as an eigenvector but where $\begin{pmatrix} -2 \\ 3 \end{pmatrix}$ is not an eigenvector.

$$\begin{pmatrix} 2 & -2 \\ 3 & -3 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} -2 \\ -3 \end{pmatrix} \text{ but } \begin{pmatrix} 2 & -2 \\ 3 & -3 \end{pmatrix} \begin{pmatrix} -2 \\ 3 \end{pmatrix} = \begin{pmatrix} -10 \\ -15 \end{pmatrix}, \text{ not a scalar multiple of } \begin{pmatrix} -2 \\ 3 \end{pmatrix}.$$

U) A matrix whose minimal polynomial is equal to its characteristic polynomial.

$$\begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$$

V) A 3×3 matrix whose minimal polynomial is $(1 - \lambda)(4 - \lambda)$.

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{pmatrix}$$

W) A matrix with characteristic polynomial $\lambda^2 - 7\lambda + 12$ and minimal polynomial $\lambda - 1$.

Not possible. We have a theorem that states that a matrix's minimal polynomial divides its characteristic polynomial, and $\lambda - 1$ does not divide $\lambda^2 - 7\lambda + 12 = (\lambda - 3)(\lambda - 4)$

2. Explain what is wrong about the following definitions and correct them.

A vector v is an **eigenvector** of an operator A if there exists a scalar λ such that $Av = \lambda v$.

We must specify that it is a *nonzero* vector.

In an inner product space, a set of vectors is **orthonormal** if each pair of distinct vectors is orthogonal (so that the inner product of the pair is 0).

In addition to being orthogonal, the vectors in an orthonormal basis must be *normalized* (have norm 1).

3. Complete the following:

a) State the Diagonalization Theorem. Illustrate the theorem with an example.

A matrix A (with entries in $\mathbb{F}$) can be written $A = SDS^{-1}$, where D is a diagonal matrix and S is an invertible one (both with entries in $\mathbb{F}$) if and only if there exists a basis in $\mathbb{F}^n$ of eigenvectors of A . Moreover, in this case diagonal entries of D are the eigenvalues and the columns of S are the corresponding eigenvectors (column number k corresponds to k th diagonal entry of D).

For instance, the matrix $A = \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix}$ has eigenvalues 5 and 3 corresponding to eigenvectors $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$, respectively. By the theorem, we have $A = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 5 & 0 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}^{-1}$

b) Give four examples of conditions that are equivalent to a square matrix being invertible.

A square matrix has linearly independent columns.

A square matrix has linearly independent rows.

A matrix has nonzero determinant.

A matrix does not have 0 as an eigenvalue.

c) State the Cayley–Hamilton Theorem. Illustrate the theorem with an example.

An operator satisfies its own characteristic polynomial. That is, if an operator A has characteristic polynomial p , then $p(A) = 0$, where 0 is the zero matrix.

For instance, the characteristic polynomial of $\begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix}$ is $(5 - \lambda)(3 - \lambda) = \lambda^2 - 8\lambda + 15$, and we have

$$\begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix}^2 - 8 \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix} + 15 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 17 & 8 \\ 8 & 17 \end{pmatrix} - \begin{pmatrix} 32 & 8 \\ 8 & 32 \end{pmatrix} + \begin{pmatrix} 15 & 0 \\ 0 & 15 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

d) Correct the following theorem statement:

Upper triangular (Schur) representation (Thms. 6.1.1 & 6.1.2): Let A be the matrix of an operator on an inner product space. Then A can be represented as $A = UTU^*$, where U is a normal matrix and T is an upper triangular matrix.

In the complex case described here, U can specifically be taken as a unitary matrix, which is a more restrictive condition than a normal matrix.

4. Prove each of the following:

a) If λ is an eigenvalue of a square matrix A , then $\det(\lambda I - A) = 0$.

If λ is an eigenvalue of A , by definition we have $Av = \lambda v$. Rearranging this matrix equation, and by linearity, we also have $(\lambda I - A)v = 0$. There exists a non-zero vector v for the matrix $(\lambda I - A)$ exactly when this matrix $(\lambda I - A)$ is singular. By the Invertible Matrix Theorem, this happens exactly when $\det(\lambda I - A) = 0$.

b) Use the properties of an inner product to show that the following function $\$$ on the vector $\mathbb{R}^2$ is not an inner product: $\begin{pmatrix} a \\ b \end{pmatrix} \$ \begin{pmatrix} c \\ d \end{pmatrix} = ac - bd$.

Since $\mathbb{R}^2$ is a real vector space, a function on two vectors to its scalar field $\mathbb{R}$ is an inner product when the function is symmetric, bilinear, non-negative, and non-degenerate. The function $\$$ fails to be symmetric, since $v\$w = -(v\$w)$. It also fails to be non-negative, since $\begin{pmatrix} 1 \\ 2 \end{pmatrix} \$ \begin{pmatrix} 1 \\ 2 \end{pmatrix} = -3$. And finally, it fails to be non-degenerate, since $\begin{pmatrix} 1 \\ 1 \end{pmatrix} \$ \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 0$, that is, there is a non-zero vector x such that $x\$x = 0$.

c) If A is a real 2×2 matrix with eigenvalues 1 and -1 , then $A^2 = I$.

Since A has two eigenvalues, they each have algebraic multiplicity at least 1. Since it is 2×2 , they each have algebraic multiplicity exactly 1. (The characteristic polynomial of A has degree 2.) Since geometric multiplicity of each eigenvalues is at least 1 (and so exactly 1 here, since $\text{GM} \leq \text{AM}$), we have that A is diagonalizable, and so in some basis B can be written as $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$. Then we have

$$A^2 = B \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} B^{-1} B \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} B^{-1} = B I B^{-1} = I$$

d) For any non-constant real polynomial p , there exists a complex matrix A where $p(A) = 0$.

Let p have degree n . By the Fundamental Theorem of Algebra, $p(\lambda)$ can be written as a product of linear factors $\prod_{1 \leq i \leq n} (\lambda - \lambda_i)$. Then any matrix that has as its eigenvalues the λ_i has $p(\lambda)$ as its characteristic polynomial, and so satisfies p by the Cayley–Hamilton theorem. Any upper triangular matrix with the λ_i along the diagonal can be taken for A .

5. Always, sometimes, or never? If it is always or never true, give a proof. If it is sometimes true, give an example where it's true and where it's false.

a) A symmetric matrix has all real eigenvalues.

This is sometimes true. It is true for real symmetric matrices (by the Spectral Theorem), so for instance the symmetric matrix $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ has real eigenvalues, namely 1 and -1 . However, for complex matrices, the requirement for real eigenvalues isn't symmetric but rather conjugate symmetric (which is to say self-adjoint). So for instance the matrix $\begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}$ is a symmetric complex matrix that does not have real eigenvalues.

b) For a nilpotent matrix N , $N^{100} = 0$ where 0 is the zero matrix.

Sometimes. It's true for every nilpotent matrix of dimension at most 100. But if you take 100th power of the dimension 101 nilpotent matrix that's a Jordan block with 0's on the diagonal, you get a matrix that's all 0's except a 1 in the upper right corner, the $a_{1,101}$ entry. (The last diver hasn't dived in yet!) And in particular, it's not the zero matrix.

c) An upper triangular matrix is diagonalizable.

Sometimes. $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ is diagonalizable, but $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ isn't.

d) A square matrix A and its transpose A^T have the same eigenvectors.

Sometimes. It's true for any symmetric matrix, for instance. But it's not true for $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$: it has the first standard basis vector as an eigenvector (along the x -axis), but its transpose has the second standard basis vector (along the y -axis) as an eigenvector.

e) Two distinct eigenspaces of an operator have a non-zero vector in common.

Never. This would mean that a given vector had two different eigenvalues, which is impossible.

f) Two distinct subspaces of a vector space have a non-zero vector in common.

Sometimes. Two distinct planes in $\mathbb{R}^3$ always share a line and all the vectors along the line. On the other hand, two distinct lines in $\mathbb{R}^3$ only ever share the zero vector.

g) The Euclidean norm of a vector v in $\mathbb{C}^n$ equals the Euclidean norm of the conjugate of v .

Always, since the Euclidean norm comes from the conjugate symmetric version of the dot product.

h) A self-adjoint matrix is symmetric.

Sometimes. A self-adjoint matrix is conjugate symmetric, so it can sometimes be symmetric, if all of its entries are real. But for instance $\begin{pmatrix} 1 & i \\ -i & 2 \end{pmatrix}$ is self-adjoint but not symmetric.

i) A self-adjoint matrix is a normal matrix.

Always. Normal means a matrix commutes with its adjoint; self-adjoint adjoint means a matrix is equal to its adjoint, and a matrix always commutes with itself: $AA = AA$.

j) A square matrix is equal to a product of elementary matrices.

Sometimes. Every invertible matrix is a product of elementary matrices (this is one of our important matrix decompositions). But non-invertible matrices are not products of elementary matrices, since elementary matrices are invertible and a product of invertible matrices is invertible.

k) The sum of two diagonalizable matrices is diagonalizable.

Sometimes. The sum of any two $n \times n$ diagonal matrices is also diagonal. But for instance

$$\begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} -1 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

The first two matrices are diagonalizable, since they each have two distinct eigenvalues (read off from the diagonal, since they are upper triangular); and so have GM=AM for both, since they are 2×2 . But their sum is not diagonalizable, since its only eigenvalues is 1 and has $1=GM < AM=2$.

l) The sum of two orthogonal matrices is orthogonal.

Sometimes. The following sum fails:

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$$

since the last matrix is not orthogonal (its columns do not form an orthonormal basis). However, this other sum works:

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} -1/2 & \sqrt{3}/2 \\ -\sqrt{3}/2 & -1/2 \end{pmatrix} = \begin{pmatrix} 1/2 & -\sqrt{3}/2 \\ \sqrt{3}/2 & 1/2 \end{pmatrix}$$

The trick is that you need for the sum of the first columns to still be a unit vector. Idea: use them to make an equilateral triangle.

m) If a matrix A is real, then $A - iI$ is invertible.

Sometimes. It's true for $A = I$, since

$$A - iI = I - iI = (1 + i)I$$

which has nonzero determinant $(1+i)^n$ and is therefore invertible.

On the other hand, if we let A be the 2×2 the rotation matrix

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} - i \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} i & -1 \\ 1 & i \end{pmatrix}$$

We see that the columns are linearly dependent, and so for this example $A - iI$ is not invertible.

n) If $u \cdot v = 0$ and $v \cdot w = 0$, then $w \cdot u = 0$.

Sometimes. It's true if we let the three vectors be the standard basis vectors in $\mathbb{R}^3$.

It's false if we let u and v be a pair of orthogonal vectors and let w be any nonzero multiple of u .

o) A given square matrix is either invertible or nilpotent.

Sometimes. It's true if we take any invertible or nilpotent matrix. But there are matrices that are neither. For instance, $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ is not invertible, since it has 0 determinant. But it is also not nilpotent, since it has an eigenvector with a nonzero eigenvalue. (Such a vector will never be sent to 0 under iteration of the matrix, and so no power of the matrix is the zero matrix.)

p) Two matrices with the same characteristic polynomial are equal.

Sometimes. For true, take any two equal matrices; they have the same characteristic polynomial and are equal. But there are many pairs of matrices we could take that have the same characteristic polynomial but are not equal. For instance, two similar but non-equal matrices still have the same characteristic polynomial. Or take either pair

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \text{ and } \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

or

$$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \text{ and } \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

For each pair, the matrices are not even similar, but they have the same characteristic polynomial.

q) Two matrices with the same characteristic polynomial have the same determinant.

Always. By one of our main theorems, we have that the product of a matrix's eigenvalues is equal to the determinant. But the product of the eigenvalues can be read off of the characteristic polynomial—it's equal to $(\pm)$ the constant term (depending on whether the polynomial is even or odd degree).

r) Orthogonal projection to a plane in $\mathbb{R}^3$ is represented by an orthogonal matrix.

Never. Every such projection is non-invertible, but every orthogonal matrix is invertible (its inverse is its transpose).

s) A diagonal matrix is nilpotent.

Sometimes. $\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$ is diagonal and nilpotent; $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ is diagonal and not nilpotent.

6. a) Compute the eigenvalues of the following matrix in $M_{2 \times 2}^{\mathbb{Z}/5\mathbb{Z}}$, as well as a basis for each eigenspace:

$$\begin{pmatrix} 1 & 4 \\ 4 & 1 \end{pmatrix}$$

We have the characteristic polynomial $(1 - \lambda)(1 - \lambda) - 1 = \lambda^2 - 2\lambda = \lambda(\lambda - 2)$

So we have that the eigenvalues of the matrix are 0 and 2. Computing $\text{Nul}(A - \lambda I)$ for each of these values yields:

$$\begin{aligned} \begin{pmatrix} 1 & 4 \\ 4 & 1 \end{pmatrix} &\xrightarrow{-R_2} \begin{pmatrix} 1 & 4 \\ 1 & 4 \end{pmatrix} \xrightarrow{R_2 - R_1} \begin{pmatrix} 1 & 4 \\ 0 & 0 \end{pmatrix} \\ \begin{pmatrix} -1 & 4 \\ 4 & -1 \end{pmatrix} &= \begin{pmatrix} 4 & 4 \\ 4 & 4 \end{pmatrix} \xrightarrow{R_2 - R_1} \begin{pmatrix} 4 & 4 \\ 0 & 0 \end{pmatrix} \end{aligned}$$

So we have $\lambda = 0$ with the vector $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ as a basis for its eigenspace; and $\lambda = 2$ with the vector $\begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \\ 4 \end{pmatrix}$ as a basis for its eigenspace.

b) Compute the eigenvalues (with multiplicity) of the following linear operator on $\mathbb{R}^3$: reflection across the plane $x + 2y + z = 0$.

This transformation is similar to (and so has the same eigenvalues as) the standard reflection transformation

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

and so has eigenvalue 1 with AM=GM=2 and eigenvalue -1 with AM=GM=1.

c) Let D be the linear operator acting on the real vector space of real differentiable functions in a single variable x . For each of the following values λ , give an example of a vector v such that v is an eigenvector under D having eigenvalue λ : 1, 2, 0, 37.

For each we can take $e^{\lambda x}$. Then $D(e^{\lambda x}) = \lambda e^{\lambda x}$, as desired. Note that in the case of $\lambda = 0$, this yields the nonzero constant function 1, which has derivative 0. $D(1) = 0 \cdot 1$.

d) Continuing from part c, give an example of (i) a vector that is not an eigenvector for D and (ii) a vector that is not an eigenvector for D but that is an eigenvector for D^2 .

(i) The function x^2 is a real differentiable function but is not an eigenvector under D , since $D(x^2) = 2x$, and $2x$ is not a scalar multiple of x^2 .

(ii) The function $\sin(x)$ is a real differentiable function. It is not an eigenvector under D since $D(\sin(x)) = \cos(x)$ and $\cos(x)$ is not a scalar multiple of $\sin(x)$. However, $D^2(\sin(x)) = -\sin(x)$, which is a scalar multiple of $\sin(x)$, and so $\sin(x)$ is an eigenvector for D^2 with eigenvalue -1 .

7. a) Find the singular values of each of the following matrices:

$$\begin{pmatrix} 1 & 2 \\ 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 4 & 0 \\ 0 & 10 \end{pmatrix}$$

Since these are real matrices, for each we just need to compute the square roots of the eigenvalues of $A^T A$. Note that in the second case this is easily seen to yield 4 and 10. In the former case we have:

$$\begin{pmatrix} 1 & 0 & 1 \\ 2 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 2 & 2 \\ 2 & 5 \end{pmatrix}$$

This last matrix has eigenvalues 6 and 1, so the singular values of the original matrix are $\sqrt{6}$ and 1.

8. Compute the 10th power of each of the following matrices. You do not need to fully simplify the entries of the resulting matrix; it is fine if an entry is something like $5 \cdot 4^{10}$. (Hint: use either the Diagonalization Theorem or the $D + N$ decomposition.) Also, compute the determinants and eigenvalues of the resulting matrices.

$$\begin{pmatrix} 1 & 3 \\ 3 & 1 \end{pmatrix}, \begin{pmatrix} -3 & 4 \\ -9 & 9 \end{pmatrix}$$

For more details, see the “Taking powers of 2×2 matrices” handout.

First compute the diagonalization of the first matrix. Computing its eigenvalues and eigenvectors yields

$$\begin{pmatrix} 1 & 3 \\ 3 & 1 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 4 & 0 \\ 0 & -2 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}^{-1}$$

Then for the 10th power we have

$$\begin{aligned} \begin{pmatrix} 1 & 3 \\ 3 & 1 \end{pmatrix}^{10} &= \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 4 & 0 \\ 0 & -2 \end{pmatrix}^{10} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 4^{10} & 0 \\ 0 & (-2)^{10} \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}^{-1} = \\ \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 4^{10} & 0 \\ 0 & (-2)^{10} \end{pmatrix}^{\frac{1}{2}} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} &= \frac{1}{2} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 4^{10} & 4^{10} \\ -(-2)^{10} & (-2)^{10} \end{pmatrix} = \\ \frac{1}{2} \begin{pmatrix} 4^{10} + 2^{10} & 4^{10} - 2^{10} \\ 4^{10} - 2^{10} & 4^{10} + 2^{10} \end{pmatrix} \end{aligned}$$

The second matrix $\begin{pmatrix} -3 & 4 \\ -9 & 9 \end{pmatrix}$ has a unique eigenvalue of 3, with corresponding eigenvector $\begin{pmatrix} 2 \\ 3 \end{pmatrix}$. The eigenvalue has algebraic multiplicity 2 but geometric multiplicity 1, and so the matrix is not diagonalizable. However, it is similar to the matrix $\begin{pmatrix} 3 & 1 \\ 0 & 3 \end{pmatrix}$. To find the second vector for our change of basis matrix, we solve:

$$\begin{pmatrix} -3 & 4 \\ -9 & 9 \end{pmatrix} \begin{pmatrix} c \\ d \end{pmatrix} = \begin{pmatrix} 3c + 2 \\ 3d + 3 \end{pmatrix}$$

This yields the condition $-c + 2d = 1$, so we may take for instance $v_2 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ along with $v_1 = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$ (the eigenvector) in our change of basis formula:

$$\begin{pmatrix} -3 & 4 \\ -9 & 9 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 3 & 2 \end{pmatrix} \begin{pmatrix} 3 & 1 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 3 & 2 \end{pmatrix}^{-1}$$

It is now enough to compute $\begin{pmatrix} 3 & 1 \\ 0 & 3 \end{pmatrix}^{10}$ and plug the result into the change of basis formula. Applying the $D + N$ decomposition and the binomial theorem, we have $\begin{pmatrix} 3 & 1 \\ 0 & 3 \end{pmatrix}^{10} = \left(\begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix} + \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \right)^{10} = \begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix}^{10} + 10 \begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix}^9 \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 3^{10} & 10 \cdot 3^9 \\ 0 & 3^{10} \end{pmatrix}$

And substituting back into the change of basis formula, we have that

$$\begin{aligned} \begin{pmatrix} -3 & 4 \\ -9 & 9 \end{pmatrix}^{10} &= \begin{pmatrix} 2 & 1 \\ 3 & 2 \end{pmatrix} \begin{pmatrix} 3 & 1 \\ 0 & 3 \end{pmatrix}^{10} \begin{pmatrix} 2 & 1 \\ 3 & 2 \end{pmatrix}^{-1} = \begin{pmatrix} 2 & 1 \\ 3 & 2 \end{pmatrix} \begin{pmatrix} 3^{10} & 10 \cdot 3^9 \\ 0 & 3^{10} \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 3 & 2 \end{pmatrix}^{-1} = \\ \begin{pmatrix} 3^{10} - 60 \cdot 3^9 & 40 \cdot 3^9 \\ -90 \cdot 3^9 & 3^{10} + 60 \cdot 3^9 \end{pmatrix} \end{aligned}$$

9. Compute a QR decomposition of the following matrix:

$$\begin{pmatrix} 3 & 1 \\ 4 & 1 \end{pmatrix}$$

We perform Gram–Schmidt on the columns by applying elementary matrices on the right:

$$\begin{pmatrix} 3 & 1 \\ 4 & 1 \end{pmatrix} \begin{pmatrix} 1/5 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & -7/25 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 5 \end{pmatrix} = \begin{pmatrix} 3/5 & 4/5 \\ 4/5 & -3/5 \end{pmatrix}$$

So we have the QR decomposition

$$\begin{pmatrix} 3 & 1 \\ 4 & 1 \end{pmatrix} = \begin{pmatrix} 3/5 & 4/5 \\ 4/5 & -3/5 \end{pmatrix} \begin{pmatrix} 5 & 7/5 \\ 0 & 1/5 \end{pmatrix}$$

10. a) Find all vectors in $\mathbb{R}^4$ that are orthogonal to both of the following vectors:

$$\begin{pmatrix} 2 \\ 0 \\ 4 \\ 0 \end{pmatrix} \quad \begin{pmatrix} 0 \\ 1 \\ 1 \\ 1 \end{pmatrix}$$

This is equivalent to finding the solution set of

$$\begin{aligned} 2w + 0x + 4y + 0z &= 0 \\ 0w + x + y + z &= 0 \end{aligned}$$

which is

$$\vec{x} = \begin{pmatrix} w \\ x \\ y \\ z \end{pmatrix} = y \begin{pmatrix} -2 \\ -1 \\ 1 \\ 0 \end{pmatrix} + z \begin{pmatrix} 0 \\ -1 \\ 0 \\ 1 \end{pmatrix}$$

b) Find a basis for the subspace in $M_{2 \times 2}^{\mathbb{R}}$ that is orthogonal to the subspace of the symmetric matrices (using the standard inner product). (Looking back at problem 1(I) might help.)

The subspace of real symmetric 2×2 matrices is 3 dimensional, and the vector space of all real 2×2 matrices is 4 dimensional. So we expect to have a 1 dimensional subspace that's orthogonal to the symmetric ones.

We could write the basis for the 2×2 symmetric matrices as column vectors and then compute a nullspace, like in part (a):

$$\begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \\ 0 \end{pmatrix}$$

It also suffices to see that the matrix

$$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

is orthogonal to those three basis vectors, and so spans the orthogonal subspace. Note that this subspace is exactly the antisymmetric 2×2 real matrices!